Davide Ermellino

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Education

Eötvös Loránd University (ELTE) Sep 2027 – Jul 2028
M.Sc. Data Science, EIT Digital Master School (double degree, year 2) *Budapest, HU*
- Coursework: Deep Network Development, Network Science, Stream Mining, Image Processing, Open-source Technologies for Data Science

Universidad Politécnica de Madrid (UPM) Sep 2026 – Jul 2027
M.Sc. Data Science, EIT Digital Master School (double degree) *Madrid, ES*
- Minor in Innovation & Entrepreneurship
- Coursework: Statistical Data Analysis, Data Processes, Cloud Computing and Big Data Ecosystems Design, Generative AI and Language Models, Intelligent Systems, Information Retrieval

University of Cagliari Oct 2023 – Jul 2026
B.Sc. Applied Computer Science and Data Analytics, 110/110 cum laude *Cagliari, IT*
- Coursework: Computational Statistics, Machine Learning, Data Structures and Algorithms, Deep Learning, Big Data

Aulab Jun 2023 – Sep 2023
Full-Stack Web Developer Bootcamp (400 h) and React Masterclass (50 h) *Bari, IT*
- Coursework: JavaScript, React, PHP/Laravel, HTML/CSS, Agile/Scrum

Experience

Intelligent Systems Group (GSI), Universidad Politécnica de Madrid Oct 2026 – Present
Research Assistant (part-time) *Madrid, ES*
- Work on HumAMI, a Horizon Europe project on agentic and edge AI for physical products, such as industrial machines and robots.
- Develop agent-based simulation and multi-agent system components in Python.

RE/MAX Apr 2026 – Jul 2026
Data Scientist
- Built a dataset of all 308 Portuguese municipalities from demographic and geospatial sources as input for real estate valuation models, plus an interactive D3.js dashboard for stakeholders.

University of Cagliari Jan 2026 – Jun 2026
Undergraduate Student Researcher *Cagliari, IT*
- Designed a validation and refinement loop for an LLM agent that writes Unity C# code, raising the compilation rate by 53% over the unvalidated baseline.
- Built a ground-truth dataset for the Planner and Executor agents of the XAGE pipeline and scored generated code with CodeBLEU and template precision/recall.
- Tested the scripts in VR scenes in Viroo. Full abstract at SIREM; full paper to be published in JIMTLT.

Universidad de Las Palmas de Gran Canaria Aug 2025 – Jan 2026
Undergraduate Student Researcher *Las Palmas, ES*
- Built a runner bib recognition system with vision-language models (VLMs) that reached 98% accuracy.
- Combined YOLO detection with VLMs across several camera views, cutting manual runner identification time by 80%.

University of Cagliari Oct 2024 – Feb 2025
Research Assistant *Cagliari, IT*
- Built a JavaScript GUI for a research tool on automatic citation, and wrote Python motion algorithms for a robot that cuts stains out of pizza boxes.

Projects

Energy Fraud Detection with Deep Learning Python, PyTorch
- Built a fraud and anomaly detection pipeline on smart meter data: LSTM autoencoders turn five time series per meter (consumption, outages, alarms, maintenance, contract data) into embeddings, which DNN and tree-based classifiers label as regular, anomalous or fraudulent.
- On a heavily imbalanced dataset, a soft-voting DNN ensemble reached 0.57 weighted recall against 0.03 for the tree-based models.

Reinforcement Learning with PandaGym Python, TensorFlow
- Implemented several RL algorithms to control a robotic arm on reach and push tasks, reaching over 95% success rate.

Real-time Red Light Violation Detection Python, YOLO, OpenCV
- Built a real-time pipeline that reads the licence plates of cars running red lights, cutting manual monitoring time and cost by 70%.

Honors and Awards

Semifinal winner, EIT Digital Master School Business Challenge (Oct 2026): handled the team's financial plan for the Wayyo case.

LeadTheFuture mentee: selected by the STEM mentorship non-profit for Italian students (acceptance rate under 15%).

Skills

Programming: Python, C, R, SQL, JavaScript

ML & Data: PyTorch, TensorFlow, scikit-learn, LangChain, OpenCV, D3.js, PostgreSQL/PostGIS

Languages: Italian (native), English (C1), Spanish (A2), Hungarian (A1)